

CMSE 801 - Day 19

Object oriented programming

Nov 14, 2022

General Course Announcements

- Homework #4 is due Wednesday Nov 16 at 11:59 p.m.
- Project's deadline Sunday Dec 4 at 11:59 p.m.
- Homework #5 is due Wednesday Nov 30 at 11:59 p.m.

Questions/comments from Day 19 PCA

- Just more practice! If you could go over the point of attributes in class, just to make sure I understood it correctly, that'd be very helpful.
- The last question, something in this notebook just didn't click? I understand the goal, but arriving at the goal I do not.
- I need more help understanding what the definitions of attributes and methods are in a class.
- Could you go over creating objects part again during the class?
- The methods with double underline : * , they do not appear in the list when I press "Tab". They seem different with no underline methods, but I have no idea what they stand for. b
- Going over attributes and methods again.
- Could you explain more about part 3.3 during class? Thx!
- Can you explain the dictionary? I have no idea about it.
- Going over attributes & methods, and showing us how to properly do the last question of the pre-class.

```
In [2]: adict = {'allo': 'a', 1: 'b', 2: 'c'}  
adict[1]
```

```
Out[2]: 'b'
```

```
In [3]: text = "Because the concept of Object Oriented Programming (OOP) is very differe  
print(text.upper())
```

```
BECAUSE THE CONCEPT OF OBJECT ORIENTED PROGRAMMING (OOP) IS VERY DIFFERENT FROM  
FUNCTIONAL PROGRAMMING THAT WE WERE USING TILL NOW, THE LAST PART OF THIS PRE-CL  
ASS ASSIGNMENT WILL BE REPEATED IN THE NEXT PRE-CLASS ASSIGNMENT. IT WAS ADDED T  
O THIS ASSIGNMENT SO YOU COULD LOOK IT OVER THE MATERIAL IN ADVANCE AS THERE IS  
ONLY ONE DAY BETWEEN CLASSES.
```

```
In [4]: print(text.split(" "))
```

```
['Because', 'the', 'concept', 'of', 'Object', 'Oriented', 'Programming', '(OO
```

```
P)', 'is', 'very', 'different', 'from', 'functional', 'programming', 'that', 'w  
e', 'were', 'using', 'till', 'now', 'the', 'last', 'part', 'of', 'this', 'pre-c  
lass', 'assignment', 'will', 'be', 'repeated', 'in', 'the', 'next', 'pre-class',  
'assignment.', 'It', 'was', 'added', 'to', 'this', 'assignment', 'so', 'you', 'c  
ould', 'look', 'it', 'over', 'the', 'material', 'in', 'advance', 'as', 'there',  
'is', 'only', 'one', 'day', 'between', 'classes.']
```

```
In [5]: print(text.upper().split(" ")[[:5]])  
  
['BECAUSE', 'THE', 'CONCEPT', 'OF', 'OBJECT']
```

```
In [6]: class Athlete():  
    def __init__(self, name, sport, next_training, next_competition='NA'):  
        ''' Initialize the object Building. '''  
        self.name = name  
        self.sport = sport  
        self.next_training = next_training  
        self.next_competition = next_competition  
        return  
    def set_next_competition(self, next_competition):  
        ''' Set athlete's name. '''  
        self.next_competition = next_competition  
        return  
    def get_sport(self):  
        return self.sport  
    def get_info(self):  
        s = 'Name: '+self.name+' Sport: '+self.sport+' N.T.: '+self.next_trainin  
        return s
```

```
In [7]: bob = Athlete('Bob', 'Bobsleigh', 'Nov 20')  
print(type(bob))  
print(bob.sport)  
print(bob.next_competition)  
bob.set_next_competition('Nov 29')  
print(bob.next_competition)  
print(bob.get_sport())  
print(bob.get_info())  
  
<class '__main__.Athlete'>  
Bobsleigh  
NA  
Nov 29  
Bobsleigh  
Name: Bob Sport: Bobsleigh N.T.: Nov 20 N.C.: Nov 29
```

```
In [8]: bolt = Athlete('Usain Bolt', 'Track and field', 'NA', 'Retired')  
print(bolt.get_info())  
  
Name: Usain Bolt Sport: Track and field N.T.: NA N.C.: Retired
```

```
In [9]: class Animal():  
    def __init__(self, kind, name, to_zoo=''):  
        ''' Initialize the object Building. '''  
        self.kind = kind  
        self.name = name  
        self.brought_to_zoo = to_zoo  
        return  
  
    def set_name(self, name):  
        ''' Set animal's name. '''  
        self.name = name
```

```
        return

def set_kind(self, kind):
    ''' Set animal's kind. '''
    self.kind = kind
    return

def set_arrive_to_zoo(self, to_zoo):
    ''' Set animal's date delivered to the ZOO. '''
    self.brought_to_zoo = to_zoo

def get_name(self):
    ''' Return animal's name. '''
    return self.name

def get_kind(self):
    ''' Return animal's kind. '''
    return self.kind

def get_arrive_to_zoo(self, to_zoo):
    ''' Return animal's date delivered to the ZOO. '''
    return self.brought_to_zoo
```